



Full Length Article

Graphene-doped PEDOT:PSS nanocomposite thin films fabricated by conventional and substrate vibration-assisted spray coating (SVASC)



Firuze Soltani-kordshuli, Fatemeh Zabihi, Morteza Eslamian *

University of Michigan-Shanghai Jiao Tong University Joint Institute, Shanghai 200240, China

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ABSTRACT

In this paper, successful fabrication of highly conductive transparent graphene-doped PEDOT:PSS composite thin films is reported for the first time, using conventional and substrate vibration-assisted ultrasonic spray coating (SVASC). To suppress the challenges associated with spraying of the precursor solution containing graphene, graphene sheets were broken by sonication and were uniformly dispersed and stabilized in PEDOT:PSS aqueous solution using isopropyl alcohol (IPA). The mechanism of dispersion of graphene in PEDOT:PSS aqueous solution using IPA is elucidated. The maximum electrical conductivity of 298 S.cm^{-1} was obtained for a graphene-doped PEDOT:PSS thin film, which compared to the pristine PEDOT:PSS thin films shows a ten-fold increase, with a transparency comparable to that of the indium tin oxide (ITO)-coated glass.

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1. Introduction

PEDOT:PSS or (poly(ethylene-3,4-dioxythiophene) : poly(styrenesulfonic acid)) is a conducting transparent co-polymer widely used in thin film organic optoelectronic and photovoltaic devices, as the hole transporting layer (HTL) and even as a transparent electrode [1–3]. PEDOT:PSS can be easily processed in solution and deposited on rough conducting surfaces (i.e. indium tin oxide), as surface smoothening and protective cover layer. PEDOT is a conductive polymer, whereas PSS is non-conductive and is employed to improve the dispersion of PEDOT in an aqueous solution. PEDOT:PSS aqueous solution is a suspension of dispersed micelles comprised of PEDOT-rich cores or chains, normally surrounded by PSS-rich coatings or shells [4]. Upon casting and drying, the micelles convert to inter-linked grains or chains of conductive PEDOT entangled in a continuous PSS matrix [5].

The optoelectronic properties of PEDOT:PSS thin films, such as electrical conductivity and work function, depend on the size, morphology, nanostructure, arrangement and interlocking of PEDOT and PSS segments. The nanostructure of PEDOT:PSS may be engineered by chemical methods, such as solvent treatment and inclusion

of salts and acids [5–12], by physical methods, such as heat treatment [12,13], by doping with conductive nanomaterials such as carbon nanotubes (CNT) and graphene ([14–25]), and by mechanical methods through controlling the casting process, such as by imposing ultrasonic vibration on the substrate to refine the film structure [26–29].

Conductivity enhancement by solvent treatment is due to induction of excess ionic charges in the polymeric chains [12]. Treating the PEDOT:PSS thin film with neutral hybrids, such as zwitterions, is another approach to enhance the film conductivity [8]. Zwitterions are able to release either positive or negative charges, depending on the surrounding media. These additional charges are imposed between PEDOT and PSS and increase the conductivity. Addition of typical ionic salts to PEDOT:PSS, such as CuBr_2 , InI_3 , and CuCl_2 , or highly polar organic solvents, such as dimethyl sulfoxide (DMSO) or ethylene glycol (EG), to the precursor solution can also improve the conductivity. Ionic dissociation in aqueous media results in fragmentation of PSS chains and the release of PEDOT segments, increasing the density of electrical charges in the solution and consequently in the solid film [10]. We have recently investigated the addition of a long-chain tertiary amine, $[(\text{C}_{18}\text{H}_{37})\text{N}(\text{CH}_2\text{CH}_2\text{O})_n\text{H}(\text{CH}_2\text{CH}_2\text{O})_m\text{H}, (n+m=15)]$, abbreviated as PEGO, to PEDOT:PSS aqueous solution, where an improvement in conductivity, transmittance, and wettability was achieved [5]. Addition of PEGO to PEDOT:PSS results in positive phase separation in PEDOT:PSS film, which facilitates the formation of 3D conducting networks of PEDOT chains, resulting in conductivity enhancement.

* Corresponding author. Tel.: +86 21 3420 7249, fax: +86 21 3420 6525.

E-mail address: Morteza.Eslamian@sjtu.edu.cn, Morteza.Eslamian@gmail.com (M. Eslamian).

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Solvent and heat treatment processes are successful, but usually expensive, complex and environmentally unfavored. The present work is, therefore, focused on improving the nanostructure and conductivity of PEDOT:PSS thin films by graphene doping, where the graphene contains a small percentage of oxygen. The films are fabricated by spray coating and substrate vibration-assisted spray coating (SVASC) processes, in a single-step, fast, and scalable process. Graphene is highly conductive, where its conductivity is determined by its lattice structure and orientation, the number of stacked carbon sheets, as well as the oxygen content, given that commercial graphene usually contains a small percentage of oxygen. As the most outstanding feature, the energy level of graphene-doped PEDOT:PSS films may be tuned, making them suitable for photovoltaic applications. Also, doping organic thin films by graphene and carbon nanotubes (CNTs) results in an improvement in the film stability, owing to the low chemical activity of such carbon-based inorganic materials. Beside superior optoelectronic properties, CNT- and graphene-doped PEDOT:PSS thin films have superior mechanical properties, including the wear resistance and hardness [16].

In the following, reported results on the fabrication of graphene-doped PEDOT:PSS thin films are discussed. Yoo et al. [14] introduced PEDOT into a graphene/PSS aqueous solution allowing the resultant solution to undergo in situ polymerization. The resultant polymeric solution was then spun and annealed to make functional composite thin films, subsequently used in a dye-synthesized solar cell, where 21% of enhancement in the device power conversion efficiency was achieved compared to the cell made using un-doped PEDOT:PSS thin film. In another study, PEDOT:PSS was spin coated using graphene and graphene quantum dots as dopants [19], where a significant improvement in conductivity was observed at the expense of slight reduction in transparency. Lin et al. [17] fabricated a polymer light emitting diode using graphene-doped PEDOT:PSS. Graphene oxide was first synthesized and then mixed with PEDOT:PSS solution, followed by a thermal reduction process to reduce the graphene oxide to graphene. With increasing graphene concentration, the conductivity of the hole-injection PEDOT:PSS layer monotonically increased. Hong et al. [15] fabricated graphene-doped PEDOT:PSS thin films with light transmission of 80% and a high electro-catalytic activity. The energy conversion efficiency of the dye-synthesized solar cell made by this composite film reached 4.5%.

In all of the above mentioned works, spin coating was used as the fabrication method, which is a batch process and lab scale method. Sustainable development of the organic solar cell technology entails the application of scalable techniques, such as spray coating for fast, roll-to-roll, and low-cost fabrications of such solution-processed layers [30]. To the best of our knowledge, there is no report on the fabrication of spray-on PEDOT:PSS films doped with graphene (and CNTs). This is perhaps due to the difficulty of spraying such particulates dispersed in PEDOT:PSS solution. In this work, therefore, we report successful fabrication of PEDOT:PSS composite films by spray coating and a modified version of spray coating, i.e., ultrasonic substrate vibration-assisted spray coating (SVASC) [26–29]. Imposing controlled ultrasonic vibration on the substrate can potentially improve the uniformity, nanostructure and conductivity of PEDOT:PSS films [31].

2. Experimental procedure

2.1. Material preparation and methods

Pristine PEDOT:PSS aqueous solution, consisting of 1.3 wt.% of commercial PEDOT:PSS (0.5% PEDOT:0.8% PSS) dispersed in water with a solution conductivity of 1 S.cm^{-1} , was purchased from Sigma-Aldrich, USA. Graphene pellets composed of 7–10 layers of sheets (thickness = 4–7 nm; average sheet diameter = 50 μm), a purity of

98%, oxygen content of 1% (with no other functional groups), and electrical conductivity of 10^5 S.cm^{-1} were purchased from Hengqiu Graphene Technology Suzhou Co., Ltd. Shanghai, China. Analytical grade IPA with purity of 99.7% was purchased from Shanghai Lingfeng Chemical Reagent Co., Ltd, Shanghai, China.

Unlike Hong et al. [15] and Du et al. [32] who mixed graphene and PEDOT:PSS aqueous solution with the aid of surfactants or additives, in this study graphene was dispersed in IPA forming stable suspension without having to use surfactants [33]. To prepare a suspension of graphene in IPA, 1 g of graphene was added to 10 mL of IPA, sonicated for two hours to break the large graphene sheets. The resulting mixture was then filtered using a $0.45 \mu\text{m}$ PTFE filter. The achieved filtrate had black color after 48 hrs. Mass ratio of graphene in IPA in the filtrate solution was simply calculated to be 4.2% by measuring the weight difference between equal volumes of pure IPA and the filtrate solutions. The graphene/IPA mixture was then added to pristine PEDOT:PSS aqueous solution with a volume ratio of 1:2, respectively, and was stirred for three hours. Therefore, given that the concentration of the pristine PEDOT:PSS aqueous solution is 1.3 wt.%, it is deduced that the concentration of graphene and PEDOT:PSS solid contents in water and IPA solvents is about 2.27 wt.%.

Substrates (bare glass; 25 mm $\times$ 25 mm $\times$ 2.5 mm) were washed by detergent and deionized water in an ultrasonic bath, dried in a vacuum furnace and cleaned in an ultraviolet cleaner. Spray pyrolysis/coating equipment (Holmarc Opto-Mechatronics Pvt. Ltd., Model HO-TH-04, India) was used to prepare spray-on thin films. The machine uses an ultrasonic nozzle that generates droplets with the average size of 50 μm at the nozzle exit, with a nozzle ultrasonic vibration frequency of 40 kHz. The ultrasonically atomized droplets are carried smoothly to the substrate by carrier air at a moderate pressure of 0.3 MPa to avoid droplet scattering in the air and splashing on the substrate. A 2D traveling arm connected to the ultrasonic nozzle head controls the direction and velocity of the nozzle tip. The lateral nozzle velocity (50 mm/s in this study), the range of nozzle movement, the PEDOT:PSS solution flow rate (5 ml/min in this study), and the number of consecutive spray passes (6 spray passes in this work) were controlled and monitored by the spray coater software. The vertical distance between the nozzle tip and the substrate was set at 30 mm. An ultrasonic transducer (40 kHz) installed inside the top side of a metal box was used as the substrate holder to impose ultrasonic vibration on the glass substrates in selected experiments. The power of the ultrasonic transducer was controlled by a signal generator (Clangsonic, SONOCLG MS600, China). In order to control the substrate temperature during the spraying process, the ultrasonic transducers were kept in a water bath at 80 $^{\circ}\text{C}$. As-sprayed thin films were annealed in a vacuum furnace at 80 $^{\circ}\text{C}$ for 30 min. All samples were made at least in triplicate. Fig. 1 shows pictures of the spray coater and the ultrasonic transducer box used to impose vibration on selected samples.

2.2. Analysis and characterization

Topography images and thickness values were obtained using a confocal laser scanning microscope (CLSM, model LMS700, Zeiss, Germany) using optical and laser modes. The CLSM laser images at a field of view of 400 μm $\times$ 600 μm were used for thickness measurement. The details of the film thickness measurement using CLSM can be found in Wang et al. [5] and Wang and Eslamian, [31]. It is noted that the surface of the spray-on films is usually rough compared to the surface of spin-on films [34]. Therefore, the reported thickness of spray-on films here is the average of 20 measurements performed at various spots of the film surface. The optical images of CLSM at 20 $\times$ magnification and field of view of 700 μm $\times$ 500 μm were used to provide the surface topography of thin films. High resolution and magnification images were



(a)



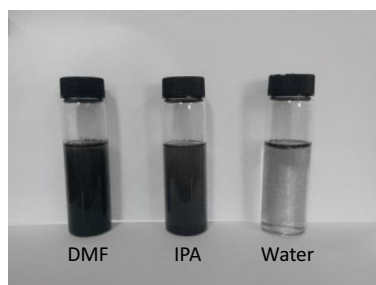
(b)

Fig. 1. Spray coating experimental setup: (a) spray coating machine including the hot plate, spray nozzle, and the data acquisition system; (b) ultrasonic vibration transducer box (left), and the signal generator (right).

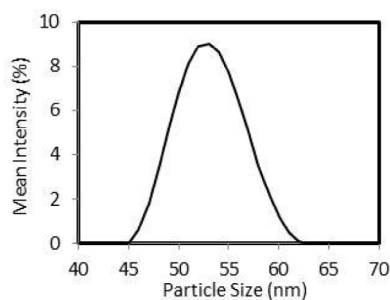
obtained using atomic force microscopy (AFM, model Dimension 3100, Veeco Instruments Inc., USA). Using AFM images, the root mean square (RMS) local surface roughness of samples was obtained. Electrical conductivity measurements were performed by linear 4-point probe technique along several 6 mm long lines on the thin film surface, and the average value is reported. The UV-Vis transmission spectra of the prepared thin films were obtained using a UV-spectrophotometer (LAMBDA 650 UV/Vis Spectrophotometer, PerkinElmer, USA). X-ray diffraction (XRD) crystallography patterns of the thin films were obtained using an XRD analyzer (D8-FABLINE, Bruker, USA). The size distribution of the suspended graphene particles dispersed in IPA was measured using a particle size analyzer (Coulter LS 130, Beckman Coulter Ltd., USA). Fourier transform infrared (FTIR, Thermo Scientific, Nicolet-50 FT-IR Spectrometer, USA) spectroscopy was performed to track the graphene in the precursor solution.

3. Results and discussion

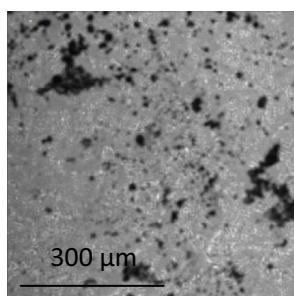
It was visually found that graphene is dispersed in IPA, more effectively than in water and dimethylformamide (DMF), making a dark uniform suspension (Fig. 2a). Fig. 1b shows the graphene particle size distribution in IPA, after filtration. The state of distribution of graphene in IPA after filtration was investigated by making a thin solid film by drop casting of graphene/IPA mixture, and characterizing the resulting film by confocal laser scanning microscope (CLSM) and atomic force microscope (AFM) (Fig. 2c and d). It is noticed that some of the graphene particulates are larger than the size of the filter used (0.45 μm PTFE), indicating some degree of particulate agglomeration due to the presence of the electrostatic forces.



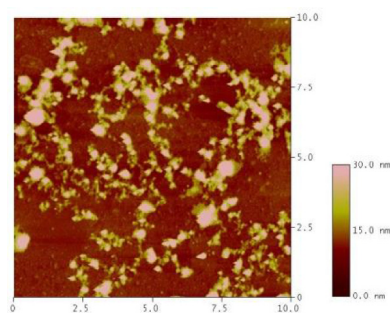
(a)



(b)



(c)



(d)

Fig. 2. (a) Picture of graphene powder dispersed in various solvents; (b) particle size distribution of graphene after filtration using 0.45 μm filter; (c) and (d) laser microscope and AFM height images of the films casted from dispersion of graphene in IPA. The AFM image is 10 μm $\times$ 10 μm .

Table 1

Experimental runs and the measured film thickness (η), roughness (ϵ) and electrical conductivity (σ) of spray-on PEDOT:PSS thin films made at the presence of IPA, except for Run 1, which is made using pristine PEDOT:PSS solution. G = graphene.

Run	Dopant	P (w)	f (kHz)	η (nm)	ϵ (nm)	σ (S·cm ⁻¹)
1 ^a	–	–	–	390	3.3	24.1
2	–	–	–	410	5.5	29.3
3	G	–	–	280	7.7	284.3
4	G	5	40	200	5.1	298.1
5	G	10	40	230	8.5	106.9

^a In Run 1, IPA was not used (pristine PEDOT:PSS).

Table 1 lists 5 experimental runs performed in this work and summarizes the results of the film thickness, roughness and conductivity of doped and un-doped PEDOT:PSS thin films, fabricated at various powers and frequencies of substrate vibration. The results shown in Table 1 will be interpreted in conjunction with the film topography images shown in Figs. 3 and 4. In Fig. 3, the effect of the addition of IPA and graphene on the morphology of the PEDOT:PSS films made using conventional spray coating (no substrate vibration imposed) is investigated. The AFM images focus on a small area of the film, while the optical images cover a much larger field of view. Given that the size of each impinging spray droplet is usually large (tens of microns), optical images can provide insight into the overall quality of the spray-on films. According to Table 1, addition of IPA to PEDOT:PSS solution results in an increase in conductivity; however the film becomes less uniform and surface defects increase, as substantiated by a considerable increase in surface roughness when IPA is added (Runs 1 and 2 of Table 1). In our previous work, treatment of PEDOT:PSS solution by low-surface tension IPA was recognized to be beneficial to liquid spreading and wetting [28]. On the other hand, the detrimental effect of IPA inclusion on the film quality observed here is attributed to a decrease in the precursor solution concentration. A decrease in solution concentration results in an increase in the film roughness, as observed before [34]. The increase in conductivity as a result of addition of IPA to the PEDOT:PSS solution may be due to the excess charges created by the organic solvent. Addition of highly conductive graphene dopants into PEDOT:PSS + IPA causes further

enhancement in conductivity (Run 3 of Table 1) and an improvement in the film intactness and mechanical strength (Fig. 3). The dispersed graphene sheets can act as a bridge or interlinking lattice in the thin film, improving the surface and inner structure of the film. As Table 1 demonstrates, the introduction of graphene to pristine PEDOT:PSS causes a 10 fold increase in the film conductivity. This is owing to the significantly high conductivity of graphene. Table 1 also demonstrates a noticeable increase in the roughness of graphene-doped PEDOT:PSS film, compared to un-doped PEDOT:PSS film, which is attributed to the existence of relatively large agglomerated dopant particulates that may protrude from the surface. Although the dopants are mixed within the mixture and tend to settle during the film formation process, still some particulates are expected to be on the film surface contributing to an increase in the film roughness.

As outlined in the introduction, we have developed and tested a modified version of spray coating process, in which controlled and low power ultrasonic vibration is imposed on the substrate to improve the film uniformity and nanostructure. The method was found quite successful when applied to pristine PEDOT:PSS solution, where a significant increase in the film conductivity and uniformity was achieved. However, a high power ultrasonic vibration was found to adversely affect the quality and intactness of the resulting solid film [28]. Our recent unpublished theoretical and experimental results show that the imposed vibration improves the precursor mixing and drying and evaporation rate, but vertical vibration has always a destabilizing effect, while lateral vibration tends to improve the film stability. Here, simultaneous vertical and lateral ultrasonic vibration is applied on composite PEDOT:PSS films, in an attempt to suppress the agglomeration of dopant particles and to increase the conductivity further and to improve the film uniformity and nanostructure. The experimental conditions and film characteristics are listed in Table 1 and the film surface topography is shown in Fig. 4. The results of Table 1 and Fig. 4 substantiate that the application of low power vibration of 5 W at a frequency of 40 kHz enhances the electrical conductivity of graphene-doped PEDOT:PSS thin films (Run 4), compared with the film made by conventional spray coating (Run 3). The film roughness also decreases substantially as a result of the imposed vibration. It should be noted

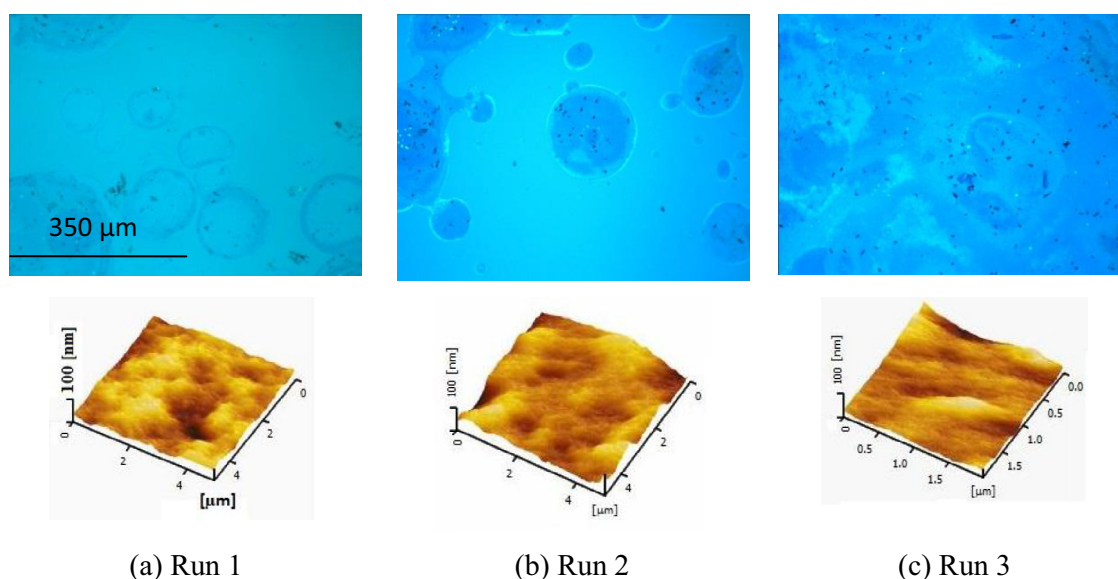


Fig. 3. Optical images (upper row) and AFM height images (lower row) showing the effect of IPA and graphene addition to pristine PEDOT:PSS solution on topography of PEDOT:PSS films made using conventional spray coating (no substrate vibration imposed). Films deposited from (a) pristine aqueous solution, (b) IPA added to the pristine solution, and (c) graphene and IPA added to the pristine solution. The field of view of the optical images is 700 μm × 500 μm. Note that the black dots are PEDOT:PSS impurities, and not graphene particles.

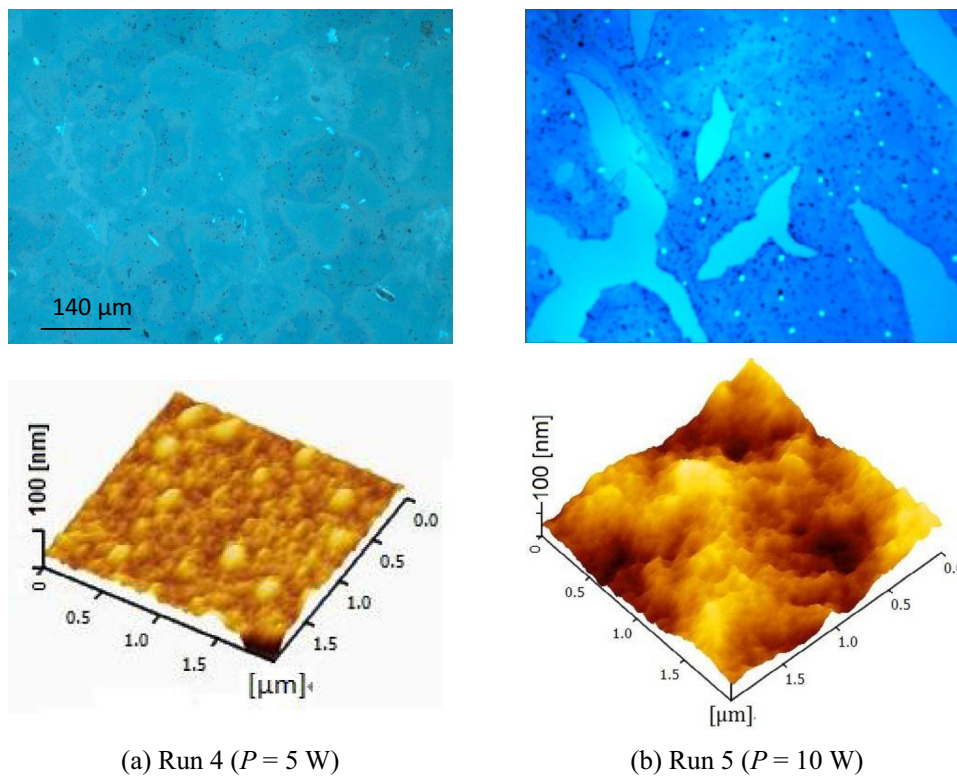


Fig. 4. Optical (upper row) and AFM (lower row) topography images of spray-on graphene-doped PEDOT:PSS thin films fabricated at two vibration powers imposed on the substrate. The field of view of the optical images is $700\ \mu\text{m} \times 500\ \mu\text{m}$.

that the improvement in conductivity and uniformity is possibly due to the uniform distribution of graphene as a result of proper imposed substrate vibration. Increasing the vibration power from 5 to 10 W deteriorates the quality and conductivity of graphene-doped PEDOT:PSS thin film, similar to what observed before for pristine PEDOT:PSS thin films [28]. The optical images show the traces of the film rupture and dewetting.

Fig. 5 shows the XRD patterns of pristine, IPA-treated, graphene-doped PEDOT:PSS films, and pure graphene. The low intensity peak of pristine PEDOT:PSS at 23° is commonly observed in the pattern of all PEDOT:PSS films [12,15]. The peak of the un-doped IPA-treated PEDOT:PSS is almost identical and overlapped with that of

the pristine PEDOT:PSS film. For graphene-PEDOT:PSS thin film, a sharp peak at 26° superimposed on PEDOT:PSS pattern clearly shows the dispersion of graphene through the PEDOT:PSS matrix, and clarifies the remarkable effect of graphene on the conductivity of PEDOT:PSS film.

Fig. 6 compares the UV-Vis transmission or transparency of undoped and doped- PEDOT:PSS thin films compared with that of the ITO-coated glass used as a standard transparent electrode in emerging solar cells. Except for the graphene-doped PEDOT:PSS film made by conventional spray coating, which shows a reduced transparency, all other films including graphene-doped PEDOT:PSS made by substrate vibration-assisted spray coating (SVASC) show a

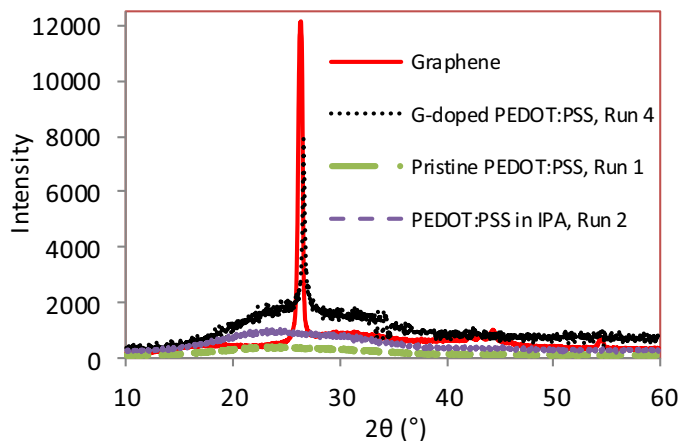


Fig. 5. XRD patterns of graphene, pristine PEDOT:PSS thin film (Run 1), PEDOT:PSS thin film made using IPA as the co-solvent (Run 2), and graphene-doped PEDOT:PSS fabricated by SVASC (Run 4).

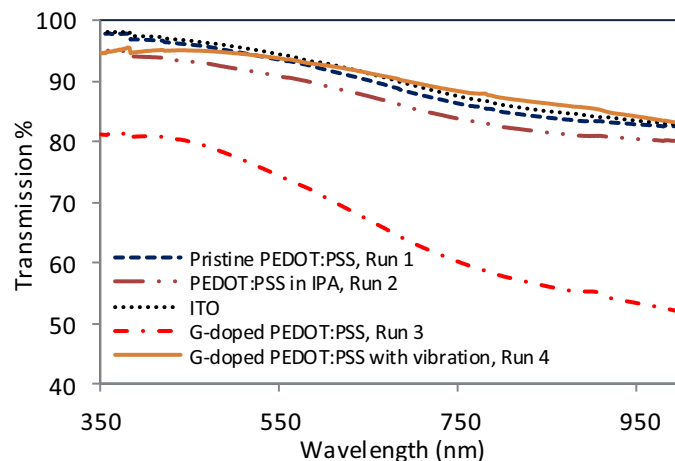


Fig. 6. UV-Vis transmission spectra of various spray-on PEDOT:PSS thin films compared with that of ITO.

Graphene (oxide) sheet

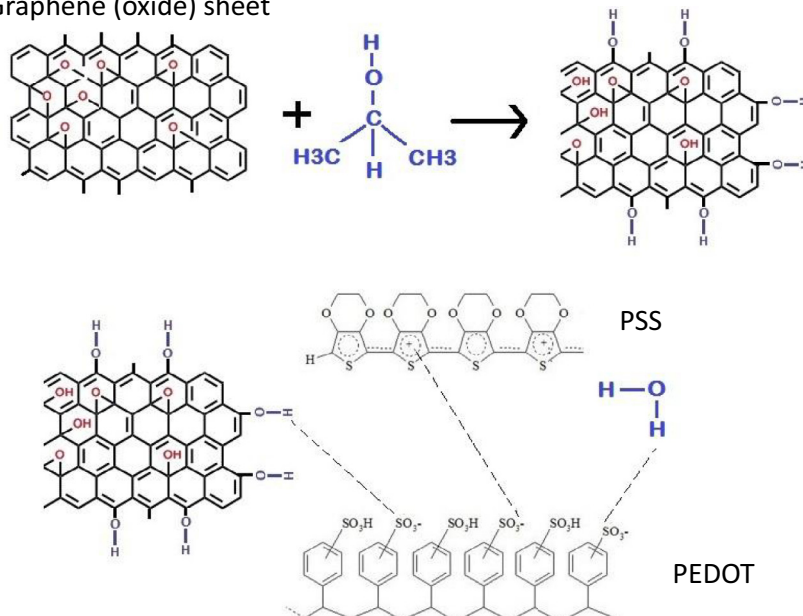


Fig. 7. Proposed chemical interactions between various groups in graphene /IPA/PEDOT:PSS/water mixture resulting in an increase in conductivity of graphene-doped PEDOT:PSS and uniform dispersion of graphene in the PEDOT:PSS matrix. The graphene sheets used in this study contain 1% oxygen.

transparency comparable to that of ITO. The trend of variation of transmission behavior of PEDOT:PSS films is consistent with those reported by others [35]. The different transmission levels result from different film structures, thicknesses and compositions. The superior transmission behavior of graphene-PEDOT:PSS film made by SVASC compared to that made by conventional spray coating is therefore attributed to lower thickness and better uniformity and distribution of graphene in the film, induced by the substrate vibration.

In the method described in section 2 for the preparation of graphene dispersion in IPA, IPA acts as a multi-functional organic medium allowing stable dispersion of graphene in IPA. Due to unsaturated carbon groups and oxygens on graphene surface, graphene can be dispersed in organic solvents better than in water [33]. Besides, full miscibility of IPA in water promotes uniformity of the precursor solution. Fig. 7 presents a proposed schematic, illustrating the mechanism of dispersion of graphene in IPA and PEDOT:PSS solution, bearing in mind that the graphene used in this work contains a small percentage of oxygen (1%). As a result of the long term blending and interaction of graphene with IPA, the $-H$ groups of IPA are attracted to oxygens on the graphene (oxide) surface. The unsaturated carbons at the edges of the broken graphene sheets are also able to bond with the $-OH$ groups in IPA. After mixing with PEDOT:PSS aqueous solution, these $-OH$ groups form hydrogen bonds with sulfurs on the hydrophilic side of PEDOT:PSS, i.e., PSS. In other words, PSS acts as the linking agent in the precursor solution. These intermolecular bonds facilitate the dispersion and suspension of graphene within the solution and contribute to enhanced conductivity of graphene-doped PEDOT:PSS. To further support the aforementioned argument, Fig. 8 shows the FTIR peaks of pristine PEDOT:PSS, IPA/graphene, and IPA/graphene/PEDOT:PSS solutions. The spikes at 1301, 1164 and 1384 cm^{-1} represent the $C-O$ stretches, attributed to the $C-O \cdots H$ bonds formed by interaction between graphene surface and $-OH$ group in IPA or water. The 1467 cm^{-1} and the strong spike at 1642 cm^{-1} are the signature of $C=C$ bond stretches in PSS and also stretching of $-OH$ groups connected to the unsaturated $C-$ groups at the edge of the fractionated

graphene sheets. The spike at 1642 cm^{-1} may be due to the highly polar $C=O$ (carboxyl) groups formed on the edge of graphene sheets, relocated by stretching from $-OH$ groups in water or IPA. The spike at 2216 cm^{-1} shows $C-O$ or $C-O_2$ stretches involved in $-OH$ of hydrogen bonding between unsaturated $C-O-$ groups and $-H$ of both organic and inorganic solvents [36–38]. This interactions are responsible for full and stable dispersion of graphene in IPA and consequently in PEDOT:PSS solution.

It is noteworthy that spray coating is a stochastic and transient process in nature [30]. Therefore, some of the film characteristics may not be 100% reproducible. From a nanoscale point of view, all samples even produced at identical conditions will have different morphology. For instance, local thickness at a particular spot of the substrate cannot be reproducible by casting methods. However, from a macroscopic point of view, considering average properties of the film, the stochastic nature of the spray can be tolerated. For in-

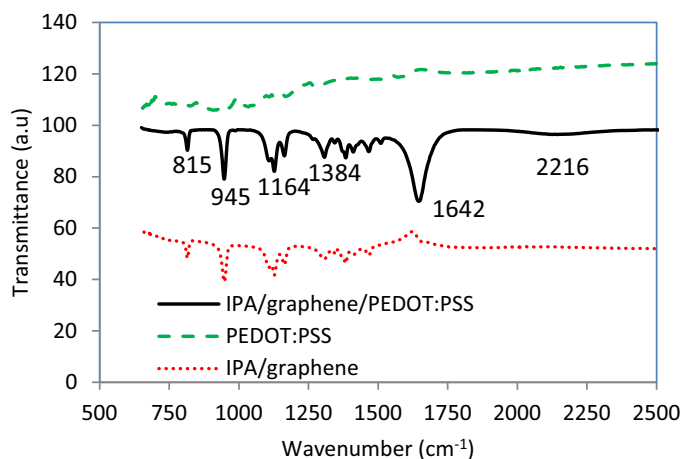


Fig. 8. FTIR spectra of pristine PEDOT:PSS, IPA/graphene/PEDOT:PSS and IPA/graphene solutions.

stance, the electrical conductivity of the spray-on films averaged over several lines across the film was found consistent.

In this work, we also attempted to fabricate CNT-doped PEDOT:PSS thin films by spray coating. However, the analyses performed on filtered solutions and thin films revealed that unlike graphene sheets which were broken into small sheets suitable for spraying, CNTs did not break into short-length tubes during the sonication process, and therefore, almost the entire CNT content was removed during the filtration process. This is attributed to high mechanical strength of the CNTs. Using a precursor containing large particulates may result in unsteady operation of the spray nozzle, or the particulates may segregate and clog the nozzle.

4. Conclusions

Highly conductive and transparent composite graphene-doped PEDOT:PSS thin films were fabricated by substrate vibration-assisted spray coating (SVASC). Up to twelve-fold increase in the conductivity of graphene-doped PEDOT:PSS thin film made by SVASC was achieved, compared to the pristine PEDOT:PSS thin film made by the conventional spray coating. Dispersion of graphene in PEDOT:PSS solution was achieved by pre-dispersion of graphene in an organic solvent (IPA). This approach eliminated the need for using surfactants, toxic solvents and heat treatment. Deposition of PEDOT:PSS thin film by conventional spray coating resulted in non-uniform and less transparent graphene-doped PEDOT:PSS thin films. Using SVASC at optimum vibration power ($P = 5$ W and $f = 40$ kHz) resulted in the formation of the most conductive, transparent and uniform graphene-doped PEDOT:PSS thin film. This is because imposing a well-controlled ultrasonic vibration on the substrate can improve wettability, spreading and merging of sprayed droplets as well as effective dispersion and distribution of graphene sheets in the liquid film. This also prevents graphene aggregation and segregation during film drying and curing. The fabrication of highly conductive graphene-doped PEDOT:PSS using scalable SVASC opens windows of opportunity for the application of composite polymeric conductive films in polymer and perovskite solar cells in large scale.

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